ROOT MEAN SQUARE IN A WIRE AND IN A GAS ON ONE PAGE

Effective value and temperature

Wherever energy goes as a square, the root mean square is the honest average, and it earns a name of its own.

THE RULE

Square, average, then take the root

An alternating current and the velocity of a gas molecule both average to zero, yet both carry energy, because heating in a resistor goes as I^2 and kinetic energy goes as v^2. Averaging the square keeps the energy and the square root restores the units:

$$x_{\mathrm{rms}} = \sqrt{\langle x^2 \rangle}, \qquad P = I_{\mathrm{rms}}^2 R \ \text{ in the wire}, \qquad \tfrac{1}{2} m v_{\mathrm{rms}}^2 = \tfrac{3}{2} kT \ \text{ in the gas}$$

In the wire the root mean square current is called the **effective value**, the steady direct current that would heat the same resistor at the same rate; older British texts call it the *virtual* value and German ones the *Effektivwert*. In the gas the same quantity, rescaled by $m/3k$, is called the **temperature**.

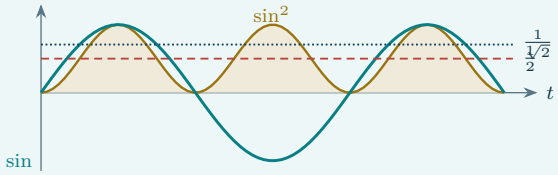
IN THE WIRE

Half the peak squared

For $I = I_0 \sin \omega t$, the square $\sin^2 \omega t = \tfrac{1}{2}(1 - \cos 2\omega t)$ swings evenly about $\tfrac{1}{2}$, so

$$I_{\mathrm{rms}} = \frac{I_0}{\sqrt{2}}, \qquad P = \tfrac{1}{2} I_0^2 R.$$

UK mains is quoted as $230\,\mathrm{V}$ effective, which means peaks of $230\sqrt{2} \approx 325\,\mathrm{V}$.



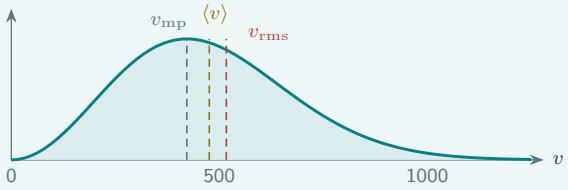
IN THE GAS

Three speeds, one temperature

The Maxwell distribution gives three typical speeds,

$$v_{\mathrm{mp}} = \sqrt{\tfrac{2kT}{m}}, \quad \langle v \rangle = \sqrt{\tfrac{8kT}{\pi m}}, \quad v_{\mathrm{rms}} = \sqrt{\tfrac{3kT}{m}},$$

and only v_{rms} fixes the mean kinetic energy. Nitrogen at $300\,\mathrm{K}$ has 422, 476 and $517\,\mathrm{m/s}$.



SIDE BY SIDE

The same arithmetic in two settings

	alternating current	ideal gas
quantity averaging to zero	current $I(t)$ over a cycle	velocity $\vec{v}$ over the molecules
energy that goes as a square	heating $I^2 R$	kinetic energy $\tfrac{1}{2} m v^2$
the root mean square	$I_{\mathrm{rms}} = I_0/\sqrt{2}$	$v_{\mathrm{rms}} = \sqrt{3kT/m}$
its name	effective (heating, virtual) value	temperature, through $\tfrac{3}{2} kT$
the plain average of the size	rectified mean $\tfrac{2}{\pi} I_0$	mean speed $\langle v \rangle$
ratio of RMS to that average	form factor $\tfrac{\pi}{2\sqrt{2}} = 1.111$	$\sqrt{3\pi/8} = 1.085$

The form factor is why a cheap meter that averages the rectified signal and multiplies by 1.111 reads true only for a pure sine wave, whereas a *true RMS* meter squares, averages and roots, just as the gas does for its temperature.

WHERE THE TWO MEET

A resistor's noise is its temperature

The electrons in any resistor jostle thermally, producing a fluctuating voltage whose effective value is set by temperature alone, the Johnson–Nyquist result

$$V_{\mathrm{rms}} = \sqrt{4kTR\,\Delta f}\,,$$

about $0.41\,\mu\mathrm{V}$ for $1\,\mathrm{k\Omega}$ at $300\,\mathrm{K}$ over a $10\,\mathrm{kHz}$ bandwidth, so an RMS voltmeter can serve as a thermometer.

HOW FAR THE ANALOGY GOES

Temperature is more than an RMS

The link $\tfrac{1}{2} m v_{\mathrm{rms}}^2 = \tfrac{3}{2} kT$ holds for translational motion in an ideal gas, where equipartition gives $\tfrac{1}{2} kT$ to each quadratic term in the energy; temperature itself is defined thermodynamically, through how entropy changes with energy, and stays meaningful for solids, radiation and spins, where no single RMS speed exists.

Conventions. $\langle\ \rangle$ is an average over a cycle or over the molecules, $k = 1.381 \times 10^{-23}\,\mathrm{J/K}$ is Boltzmann's constant, m the molecular mass and Δf the bandwidth of the measurement.

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